

Homework 6.4

Problem 13

Determine whether the natural cubic spline that interpolates the table

x	0	1	2	3
y	1	1	0	10

Is or is not the function

$$f(x) = \begin{cases} 1 + x - x^3 & x \in [0,1] \\ 1 - 2(x-1) - 3(x-1)^2 + 4(x-1)^3 & x \in [1,2] \\ 4(x-2) + 9(x-2)^2 - 3(x-2)^3 & x \in [2,3] \end{cases}$$

$$s_0(x) = 1 + x - x^3$$

$$s_1(x) = 1 - 2(x-1) - 3(x-1)^2 + 4(x-1)^3$$

$$s_2(x) = 4(x-2) + 9(x-2)^2 - 3(x-2)^3$$

$$f(0) = s_0(0) = 1 + 0 - 0^3 = 1$$

$$f(1) = s_0(1) = 1 + 1 - 1^3 = 1 = s_1(0) = 1 - 2(0) - 3(0)^2 + 4(0)^3$$

$$f(2) = s_1(2) = 1 - 2(1) - 3(1) + 4(1) = 0 = s_2(0) = 4(0) + 9(0)^2 - 3(0)^3$$

$$f(3) = s_2(3) = 4(1) + 9(1)^2 - 3(1)^3 = 10$$

$$f'(0) = s'_0(0) = 1 - 3x^2 = 1$$

$$f'(1) = s'_0(1) = 1 - 3(1)^2 = -2 = s'_1(1) = -2 - 6(0) + 12(0)$$

$$f'(2) = s'_1(2) = -2 - 6(1) + 12(1)^2 = 4 = s'_2(2) = 4 + 18(0) - 9(0)^2$$

$$f'(3) = s'_2(3) = 4 + 18(3) - 9(3)^2 = -23$$

$$f''(0) = s''_0(0) = -6(0) = 0$$

$$f''(1) = s''_0(1) = -6(1) = -6 = s''_1(1) = -6 + 24(0)$$

$$f''(2) = s''_1(2) = -6 + 24(1) = 18 = s''_2(2) = 18 - 18(0)$$

$$f''(3) = s''_2(3) = 18 - 18(1) = 0$$

This is the correct interpolation since this function satisfies all the conditions for a natural cubic spline for the given data points.

Problem 18

Which properties of a natural cubic spline does the following function possess, and which properties does it not possess?

$$f(x) = \begin{cases} (x+1) + (x+1)^3 & x \in [-1, 0] \\ 4 + (x-1) + (x-1)^3 & x \in (0, 1] \end{cases}$$

$$x_0 = -1, x_1 = 0, x_2 = 1$$

$$s_0(x) = (x+1) + (x+1)^3$$

$$s_1(x) = 4 + (x-1) + (x-1)^3$$

$$f(-1) = s_0(-1) = (0) + (0)^3 = 0$$

$$f(0) = s_0(0) = (1) + (1)^3 = 2$$

$$\lim_{x \rightarrow 0} s_1(x) = 4 + (-1) + (-1)^3 = 2$$

$$f(1) = s_1(1) = 4 + (0) + (0)^3 = 4$$

$$\therefore f \in C[-1, 1], f(x_i) = y_i, i = 0, 1, 2 \text{ but } s_0(0) \neq s_1(0)$$

$$f'(-1) = s'_0(-1) = 1 + 3(0)^2 = 1$$

$$f'(0) = s'_0(0) = 1 + 3(1)^2 = 4 = \lim_{x \rightarrow 0} s'_1(x) = 1 + 3(-1)^2$$

$$f'(1) = s'_1(1) = 1 + 3(0)^2 = 1$$

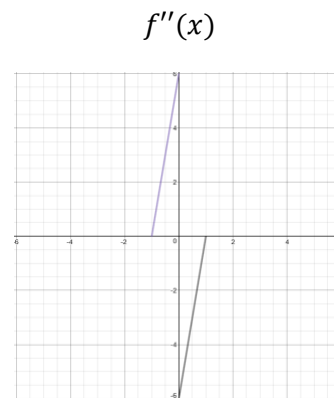
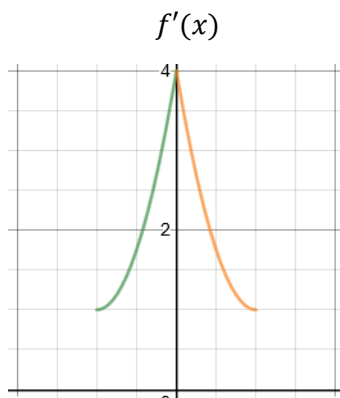
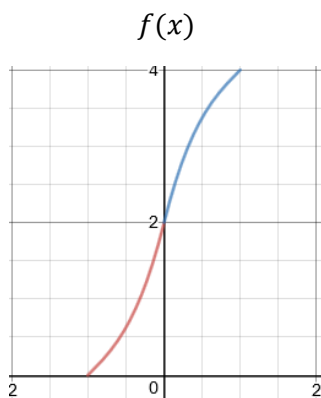
$$\therefore f \in C^1[-1, 1] \text{ but } s'_0(0) \neq s'_1(0)$$

$$f''(-1) = s''_0(-1) = 6(0) = 0$$

$$f''(0) = s''_0(0) = 6(1) = 6 \neq -6 = \lim_{x \rightarrow 0} s''_1(x) = 6(-1)$$

$$f''(1) = s''_1(1) = 6(0) = 0$$

$$\therefore f''(-1) = 0 = f''(1) \text{ but } s''_0(0) \neq s''_1(0) \text{ and } f \notin C^2[-1, 1]$$

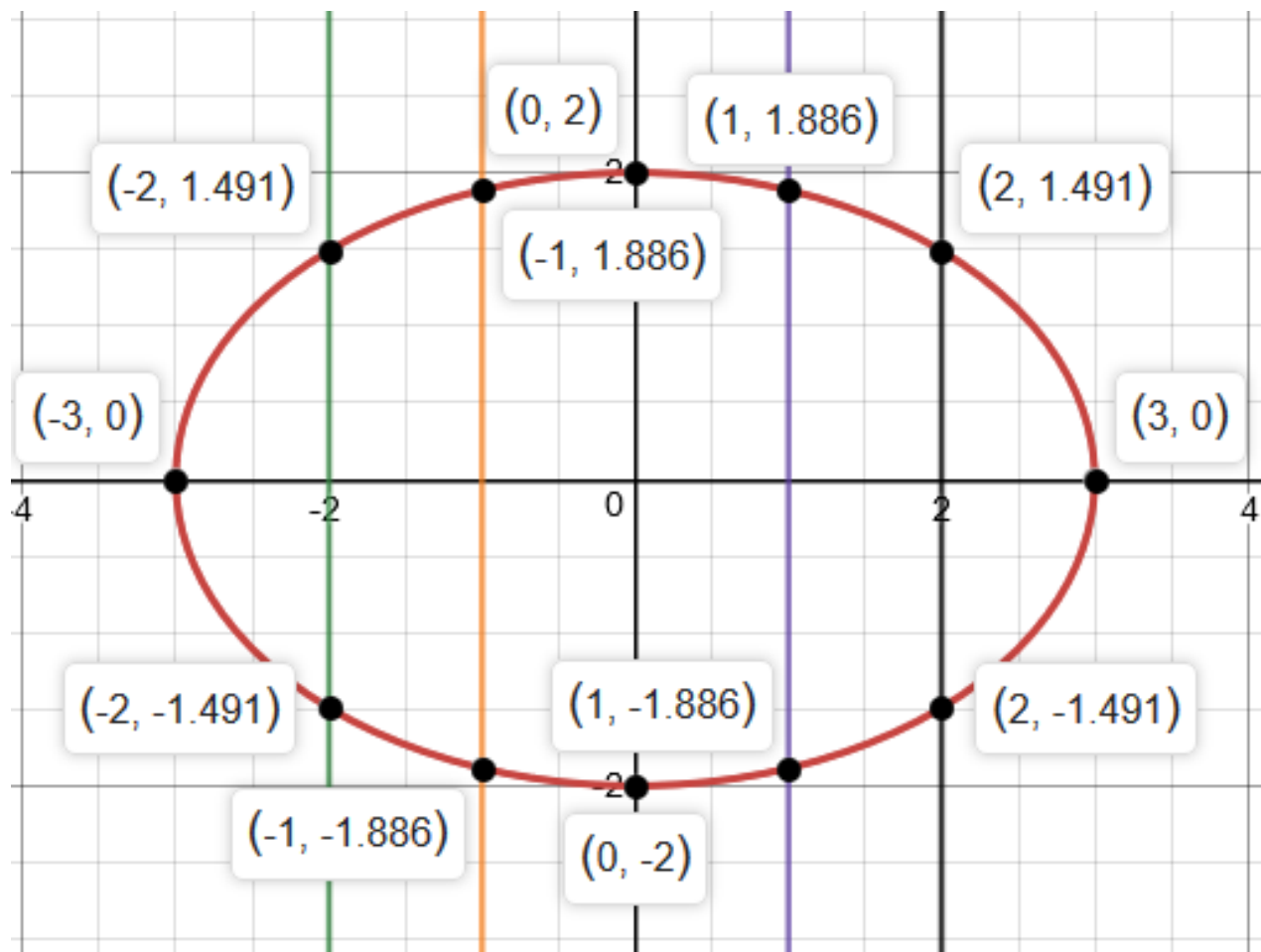


Computer Problem 6.4

Problem 1

Draw a curve, such as an oval or a spiral, on a sheet of graph paper. Select points in a more or less regular distribution along the curve and label them $t_0 = 1$, $t_1 = 2$, and so on. Read the x and y -coordinates at each selected point to obtain a table for $x(t)$ and $y(t)$. Fit these functions by spline functions S and S^* . Then the formulas $x = S(t)$ and $y = S^*(t)$ give an approximate parametric representation of the curve. Plot the resulting curves for several test cases using an automatic plotter.

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$



t	1	2	3	4	5	6	7	8	9	10	11	12	13
x	-3	-2	-1	0	1	2	3	2	1	0	-1	-2	-3
y	0	1.491	1.886	2	1.886	1.491	0	-1.491	-1.886	-2	-1.886	-1.491	0

```
>> X
X =  -3  -2  -1   0   1   2   3   2   1   0  -1  -2  -3

>> Y
Y =

Columns 1 through 10
    0  1.4910  1.8860  2.0000  1.8860  1.4910    0 -1.4910 -1.8860 -2.0000

Columns 11 through 13
-1.8860 -1.4910    0

>> T
T =   1   2   3   4   5   6   7   8   9  10  11  12  13
```

Using the following program:

```
function Spline = SP(X,T) %Using the algorythm in the book.

%Here we calculate the B vector and the H vector.
H = []; %Initialize H.
B = []; %Initialize B.
for i = 1:(length(T)-1)
    H = [H (T(i+1)-T(i))];
    B = [B 6*(X(i+1)-X(i))/H(i)];
end

%Initialize the U and V vector.
%The reason for the 0 is so that our indexing aligns correctly.
U = [0,2*(H(1)+H(2))];
V = [0,B(2)-B(1)];

%U an V vector.
for i = 2:(length(H)-1)
    %Split the calculation up to avoid errors.
    A = 2*(H(i+1)+H(i));
    P = H(i)^2;
    C = U(i);

    Un = (A-P/C);
    U = [U Un];

    %Split the calculation up to avoid errors.
    D = B(i+1)-B(i);
    E = (H(i)*V(i));
    F = U(i);

    Vn = (D-E/F);
```

```
V = [V Vn];
end

%Initialize the Z vector.
Z = [0];

%Here I had to be more crafty so that the matrix would work correctly.
j = length(T) - 2;
h = length(H);
while (j >= 1)
    Z = [Z ((V(j+1)-H(j+1)*(Z(h - j)))/U(j+1))];
    j = j - 1;
end
%End the array.
Z = [Z 0];

%Note that I ended up doing the array in reverse order to simplify the
%loop. Now I need to flip this array so that it is correct.

%This will flip the vector to be correct.
n = length(Z);
ZZ = [0];
for i = 1:(n-1)
    ZZ = [ZZ Z(n-i)];
end

%These are the values that I will be used to store the points in the cubic
%spline.
XX = [];
TT = [];

for i = 1:(length(T)-1)
    %I choose 100 so that the graph could be smoother.
    dt = (T(2)-T(1))/100;

    %This loop will work at each interval and have a t value that will go
    %between t_i and t_i+1 and will increment i once t reaches t_i+1.
    t = T(i); %Initialize t.

    while (t < T(i+1))
        %Split up the algorithm to avoid mistakes.
        A = ZZ(i)/(6*H(i))*(T(i+1)-t)^3;
        B = ZZ(i+1)/(6*H(i))*(t-T(i))^3;
        C = (X(i+1)/H(i)-(ZZ(i+1)*H(i))/6)*(t-T(i));
        D = (X(i)/H(i)-(ZZ(i)*H(i))/6)*(T(i+1)-t);

        S = A+B+C+D;

        %Add to the values of our cubic spline.
        XX = [XX S];
        TT = [TT t];
        t = t + dt; %increment t, by a little bit.
    end
end
```

```
end
```

```
%Add the end points.
```

```
TT = [TT T(length(T))];
```

```
XX = [XX X(length(X))];
```

```
%This is the plot of the graph.
```

```
%Replace V with whatever name you need.
```

```
figure('name','Graph of V vs t');
```

```
hold on
```

```
plot(TT,XX);
```

```
hold off
```

```
title('V Function of Time');
```

```
xlabel('time');
```

```
ylabel('V');
```

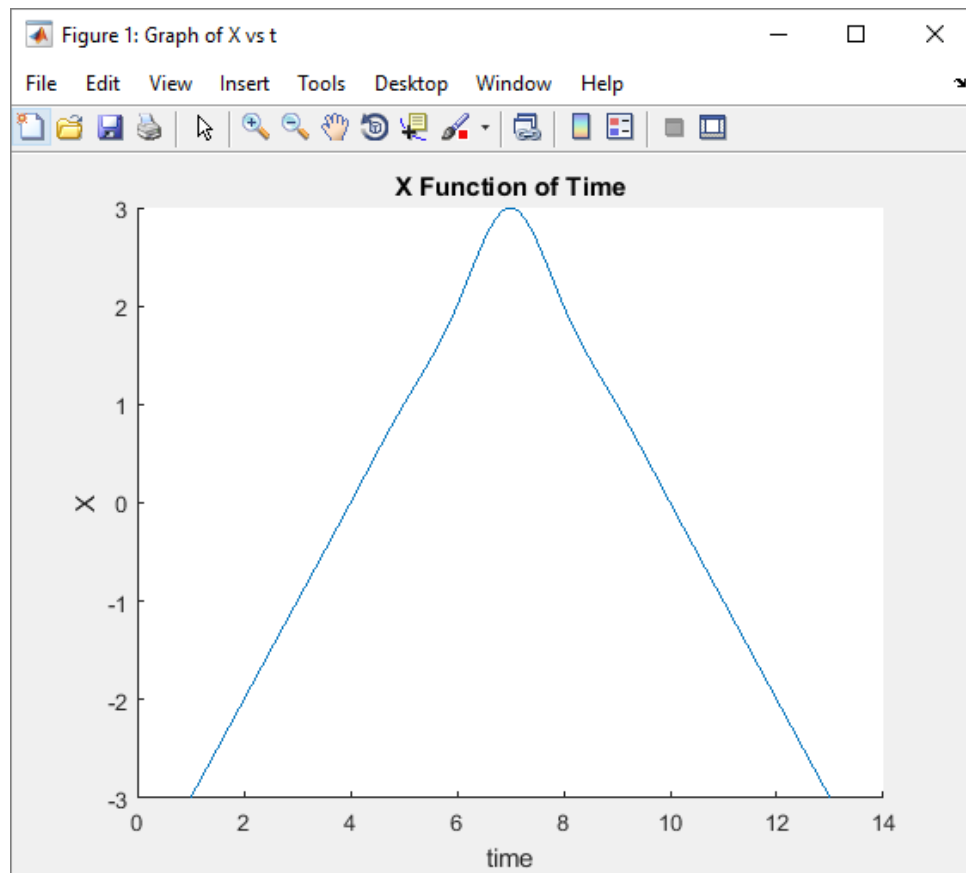
```
%Function of t values are saved since these are the values that we need for
```

```
%our parametric problem.
```

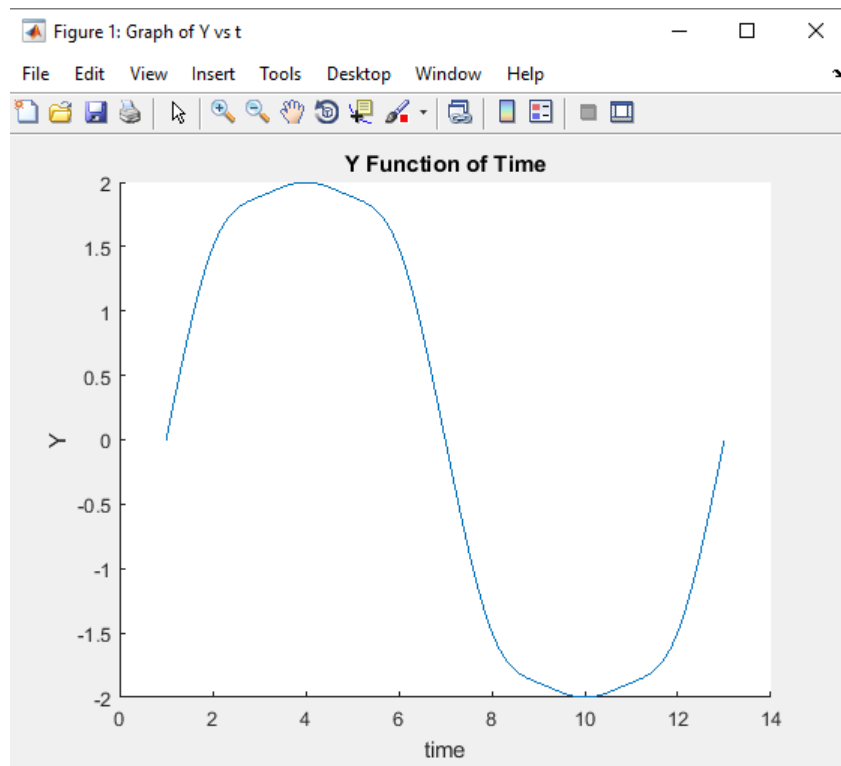
```
Spline = XX;
```

We get:

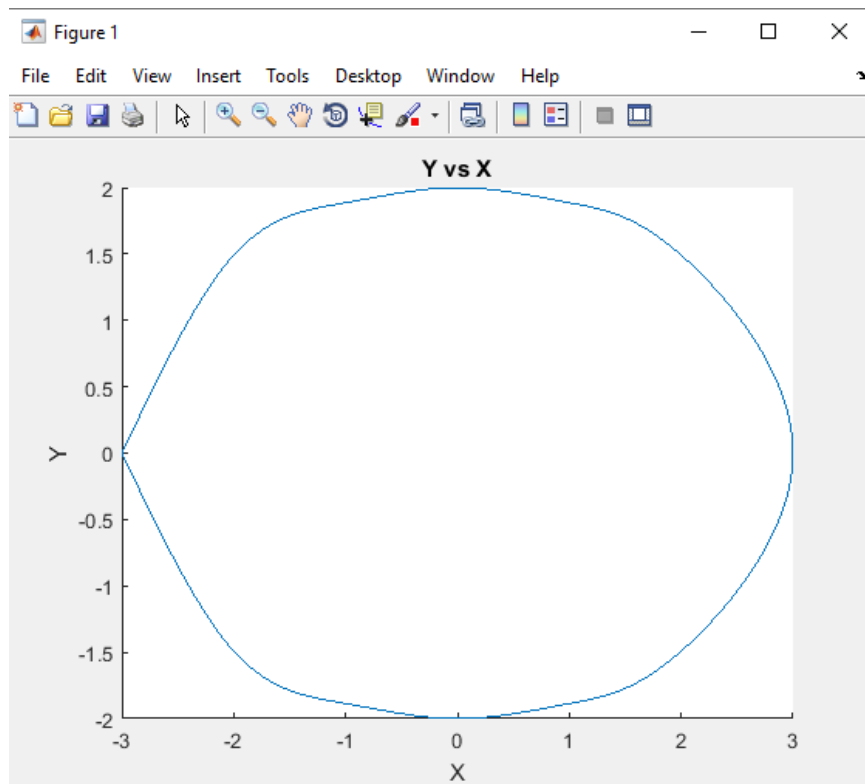
```
>> XX = SP(X,T);
```



```
>> YY = SP(Y,T);
```



```
>> plot(XX,YY)
```

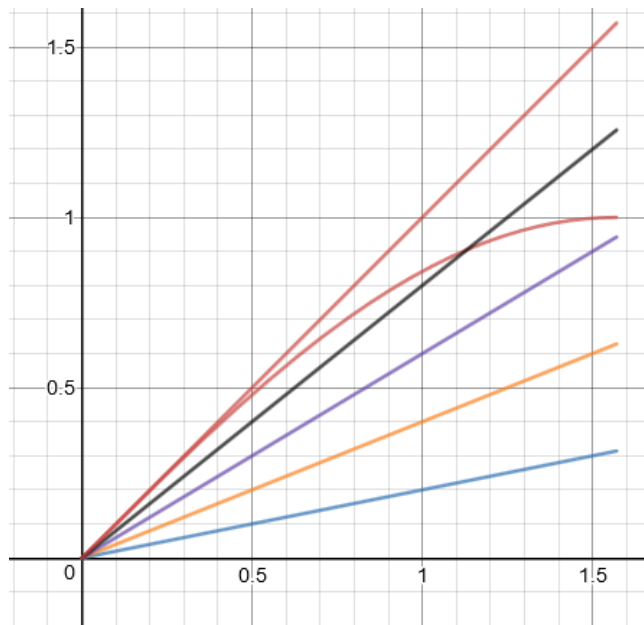


Homework 6.8

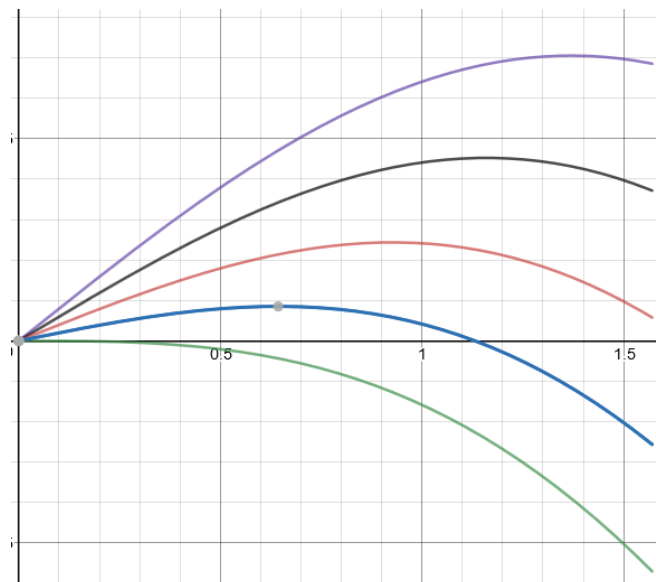
Problem 1

Find the best approximation to $\sin(x)$ by a function $u(x) = \lambda x$ on the interval $\left[0, \frac{\pi}{2}\right]$ using the supremum norm. Draw a sketch. The function $\sin(t) - \lambda t$ should have a maximum at a point $\xi \in \left(0, \frac{\pi}{2}\right)$, and it should have a minimum of the same magnitude at $\frac{\pi}{2}$. These conditions should determine ξ and λ .

Graph of $f(x)$ and $u(x)$ for $\lambda = 0.2, 0.4, 0.6, 0.8, 1.0$



Graph of $f(x) - u(x)$



From the instructions, we require that $\max\{f(x) - u(x)\} = \left| \sin\left(\frac{\pi}{2}\right) - \lambda \frac{\pi}{2} \right| = \left| 1 - \lambda \frac{\pi}{2} \right|$. Now to find the maximum, we consider the following function and use the derivatives to find the maximum.

$$\psi(x) = \sin(x) - \lambda x$$

$$\Rightarrow \psi'(x) = \cos(x) - \lambda = 0 \Rightarrow \cos(x) = \lambda \Rightarrow x = \arccos(\lambda)$$

$$\left| \psi\left(\frac{\pi}{2}\right) \right|_{\infty} = \left| \sin\left(\frac{\pi}{2}\right) - \lambda * \frac{\pi}{2} \right| = \lambda * \frac{\pi}{2} = \sin(\arccos(\lambda)) - \lambda \arccos(\lambda)$$

$$\Rightarrow 0 = \sqrt{1 - \lambda^2} - \lambda \left(\arccos(\lambda) + \left(\frac{\pi}{2}\right) \right) \Rightarrow \lambda = 0$$

$$\Rightarrow 0 = \sqrt{1 - \lambda^2} + \lambda \left(\arccos(\lambda) + \left(\frac{\pi}{2}\right) \right)$$

Using the Bisection method

```
fn =  
function_handle with value:  
@(x)sqrt(1-x^2)-x*acos(x)+(1-x*pi/2)  
>> fn(.6)  
ans = 0.301145073122095  
>> fn(.8)  
ans = -0.171437948470545  
>> B(fn,.6,.8)  
c = 0.724611353874207
```

$$\lambda = 0.724611353874207$$

Problem 2

(Continuation) Solve the preceding problem using the usual quadratic norm.

Note that since we have that the inner product in the space with this approximation needs to give us 0, we have that $\langle \psi(x), \mu x \rangle = 0$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \psi(x) \mu x dx = \mu \int_0^{\frac{\pi}{2}} (x * \sin(x) - \lambda x^2) dx = 0$$

$$\Rightarrow [-x * \cos(x) + \sin(x)]_0^{\frac{\pi}{2}} - \left[\frac{\lambda x^3}{3} \right]_0^{\frac{\pi}{2}} = 0 \Rightarrow [1 - (0)] - \left[\frac{\lambda \pi^3}{24} \right] \Rightarrow \lambda = \frac{24}{\pi^3}$$

Problem 6

Show that the notorious **Hilbert matrix** with elements

$$a_{ij} = (1 + i + j)^{-1}, 0 \leq i, j \leq n$$

Is a Gram matrix for the functions $1, x, \dots, x^{n-1}$.

Given a set of functions $\{v_1, v_2, \dots, v_n\}$ defined on $[a, b]$, the Gram matrix is as follows

$$\mathbf{G} = [G_{ij}], G_{ij} = \int_a^b v_i(x) \overline{v_j(x)} dx$$

Our set of functions is $\{1, x, x^2, \dots, x^{n-1}\}$ defined on $[0, 1]$, thus the Gram matrix is given by

$$G_{ij} = \int_0^1 x^i x^j dx = \int_0^1 x^{i+j} dx = \left[\frac{x^{i+j+1}}{i+j+1} \right]_0^1 = \frac{1}{i+j+1} = (1+i+j)^{-1}$$

Problem 15

Prove that an orthogonal set of nonzero elements is necessarily linearly independent.

Let $\Psi = \{\psi_0, \psi_1, \dots, \psi_n, \dots\}$ be a set of orthogonal vectors such that $\psi_i \neq 0$. To show that they are in fact linearly independent, we will use contradiction. Let there be a ψ_k such that it is dependent, i.e.

$$\psi_k = c_0 \psi_0 + c_1 \psi_{k_1} + \dots + c_n \psi_n + \dots \text{ (excluding } \psi_k \text{)}$$

Where $c_i \in \mathbb{F}$ and not all $c_i = 0$. Note that ψ_k is a combination of all the other vectors in Ψ . Since we have that all vectors are orthogonal, we have that

$$\langle \psi_k, \psi_i \rangle = 0, \forall i \neq k$$

Now remember that Ψ is orthogonal, thus we have that

$$\langle \psi_i, \psi_j \rangle = \begin{cases} 0 & i \neq j \\ k_i \neq 0 & i = j \end{cases}$$

This means that

$$\begin{aligned} \langle \psi_k, \psi_i \rangle &= \langle c_0 \psi_0 + c_1 \psi_{k_1} + \dots + c_n \psi_n + \dots, \psi_i \rangle \\ \Rightarrow \langle \psi_k, \psi_i \rangle &= c_0 \langle \psi_0, \psi_i \rangle + \dots + c_i \langle \psi_i, \psi_i \rangle + c_n \langle \psi_n, \psi_i \rangle + \dots \\ &\Rightarrow \langle \psi_k, \psi_i \rangle = c_i k_i = 0 \Rightarrow c_i = 0 \end{aligned}$$

Since we have that ψ_i is arbitrary, we have that $\psi_k = 0 * \psi_0 + 0 * \psi_1 + \dots + 0 * \psi_n + \dots$ which gives us that $\psi_k = 0$, which contradicts our assumption that all ψ_i are nonzero, thus we must have that Ψ is linearly independent.

Problem 16

Let A be a linear transformation on an inner-product space. Assume that A is **Self-adjoint**, which means that $\langle Af, g \rangle = \langle f, Ag \rangle, \forall f, g$. Prove that the solutions of the equation $Af = \lambda f$, corresponding to different values of λ , are mutually orthogonal.

Define the following:

$$Af = \lambda_f f$$

$$Ag = \lambda_g g$$

Where $\lambda_f \neq \lambda_g$ and $f \neq g$. Then:

$$\langle Af, g \rangle = \langle \lambda_f f, g \rangle = \lambda_f \langle f, g \rangle = \lambda_g \langle f, g \rangle = \langle f, \lambda_g g \rangle = \langle f, Ag \rangle$$

$$\lambda_f \neq \lambda_g \Rightarrow \langle f, g \rangle = 0$$

Thus, we have that these solutions are orthogonal.